Supplemental Material: A Unified Variational Principle for Branching Transport Networks

Riccardo Marchesi

University of Pavia

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S1. Coherent Wave Reflection: Transfer-Matrix Analysis

S1.1 Motivation

The main text employs an incoherent multiplicative model for wave cost accumulation (Eq. 5). In the coronary tree, where $kL \ll 1$ (the tree spans ∼1% of the pulse wavelength at the cardiac fundamental), coherent interference effects could in principle alter the total reflected power and shift the minimax crossing point. We address this concern with a full transfer-matrix calculation.

S1.2 Method

We model the coronary tree as a cascade of $G = 11$ transmission-line segments, each representing one generation. At each generation g, the segment has:

- Radius: $r_g = r_0 \cdot 2^{-g/\alpha}$
- Length: $\ell_g = 10\, r_g$ (self-similar, consistent with coronary morphometry)
- Wall thickness: $h_g = c_0\, r_g^p$ with $p = 0.77$

The **Moens–Korteweg wave speed** at generation g is:

$$c_g = \sqrt{\frac{E\, h_g}{2\rho\, r_g}} \tag{1}$$

with $E = 0.4$ MPa (typical coronary elastic modulus).

The **Womersley velocity-profile correction** F_{10} is computed using Bessel functions:

$$F_{10}(\mathrm{Wo}) = \frac{2\, J_1(\lambda)}{\lambda\, J_0(\lambda)} \tag{2}$$

where $\lambda = \mathrm{Wo} \cdot j^{3/2}$ and $\mathrm{Wo} = r_g \sqrt{\omega\rho/\mu}$ is the Womersley number.

The **characteristic impedance** and **propagation constant** of each segment are:

$$Z_{c,g} = \frac{\rho\, c_g}{\pi r_g^2}, \qquad \gamma_g = \frac{j\omega\sqrt{1 - F_{10}}}{c_g}. \tag{3}$$

The **terminal load** at generation G is a two-element Windkessel:

$$Z_{\mathrm{term}} = \frac{R_{\mathrm{term}}}{1 + j\omega R_{\mathrm{term}} C_{\mathrm{term}}} \tag{4}$$

with $R_{\mathrm{term}} = 8\mu\ell_{\mathrm{cap}}/(\pi r_G^4)$ (Poiseuille resistance of a capillary-length segment) and $C_{\mathrm{term}} = 3\pi r_G^3 \ell_G/(2Eh_G)$ (elastic compliance of the terminal vessel).

Backward recursion. Starting from $Z_{\mathrm{in}} = Z_{\mathrm{term}}$, at each generation $g = G-1, \ldots, 0$:

1. Compute $Z_{\mathrm{daughters}} = Z_{\mathrm{in}}/2$ (two daughters in parallel)
2. Apply the transmission-line input-impedance formula:

$$Z_{\mathrm{in},g} = Z_{c,g}\, \frac{Z_{\mathrm{daughters}} + Z_{c,g} \tanh(\gamma_g \ell_g)}{Z_{c,g} + Z_{\mathrm{daughters}} \tanh(\gamma_g \ell_g)} \tag{5}$$

The **total reflection coefficient** at the root is:

$$|\Gamma_{\mathrm{total}}|^2 = \left| \frac{Z_{\mathrm{in},0} - Z_{c,\mathrm{source}}}{Z_{\mathrm{in},0} + Z_{c,\mathrm{source}}} \right|^2 \tag{6}$$

S1.3 Results

Terminal-load dominance. At the impedance-matching exponent $\alpha = \alpha_w = 2.115$, where all junctions are perfectly matched, the coherent model still predicts $|\Gamma_{\text{total}}|^2(\alpha_w) = 0.026$. This residual reflection is entirely due to the resistive terminal capillary load, not junction geometry.

The physically relevant quantity for the minimax optimization is the **geometric penalty**:

$$\Delta_{\text{coherent}}(\alpha) = |\Gamma_{\text{total}}(\alpha)|^2 - |\Gamma_{\text{total}}(\alpha_w)|^2 \quad (7)$$

which isolates the α-dependent contribution from junction mismatches.

Proportionality with incoherent model. For $\alpha > 2.3$, $\Delta_{\text{coherent}}(\alpha)$ is:

1. **Monotonically increasing** — no Fabry–Pérot resonances create additional crossings
2. **Correlated with $\mathcal{C}^{\text{net}}_{\text{wave}}$ at $r = 0.91$** — the two measures are proportional

The proportionality constant (factor of ~ 12) reflects the coherent amplification expected in a $kL \ll 1$ lattice; the minimax shift relative to the incoherent model is $|\Delta\alpha^*| = 0.000 < 0.01$.

Per-harmonic analysis at $\alpha = 2.72$:

Harmonic	f (Hz)	$\lvert\Gamma_{\text{total}}\rvert^2$
1	1.2	0.813
2	2.4	0.855
3	3.6	0.890
4	4.8	0.920

The modest variation across harmonics confirms that Fabry–Pérot resonance effects are negligible in this regime (Womersley numbers $\text{Wo} < 2.3$ throughout the tree, providing viscous damping).

Frequency-averaged geometric penalty confirms a unique crossing with shift $|\Delta\alpha^*| = 0.000$ (< 0.01) relative to the incoherent model.

S1.4 Discussion

The incoherent model (Eq. 5 of the main text) measures the correct physical quantity: the geometric penalty for choosing $\alpha \neq \alpha_w$ at each junction, accumulated multiplicatively through the tree. The coherent transfer-matrix calculation confirms that this geometric penalty is the dominant α-dependent component of total wave reflection, and that the large baseline reflection is a property of the terminal load boundary condition, not of junction geometry.

Note on Architectural Consistency. In this coherent transfer-matrix calculation, the network is modeled as a uniform tree defined by the global architectural parameter α ($r_g = r_0 \cdot 2^{-g/\alpha}$), with a representative constant elastic modulus $E = 0.4$ MPa. This is done to preserve strict structural consistency with the main text, where α is treated as a global topological constraint and the transport cost $\mathcal{C}^{\text{net}}_{\text{transport}}(\alpha)$ measures the penalty of deviating from local optima. While the sub-linear wall-thickness law $h(r) \propto r^p$ inherently implies a proximodistal impedance gradient (a "chirped" lattice), evaluating the wave equations on the purely uniform baseline correctly isolates the geometric penalty $\Delta_{\text{coherent}}(\alpha)$ attributable to α alone. Fully variational optimization relaxing the global α constraint to locally varying branching ratios (ρ_g) over a chirped impedance lattice represents a natural extension of this framework.

The factor-of-12 ratio between Δ_{coherent} and $\mathcal{C}^{\text{net}}_{\text{wave}}$ reflects the coherent amplification $|\Gamma_{\text{total}}|^2 \sim G^2|\Gamma_{\text{single}}|^2$ expected in a $kL \ll 1$ lattice; the proportionality of their α-dependences ($r = 0.91$) confirms that the geometric sensitivity of the two models is consistent despite this amplitude difference.

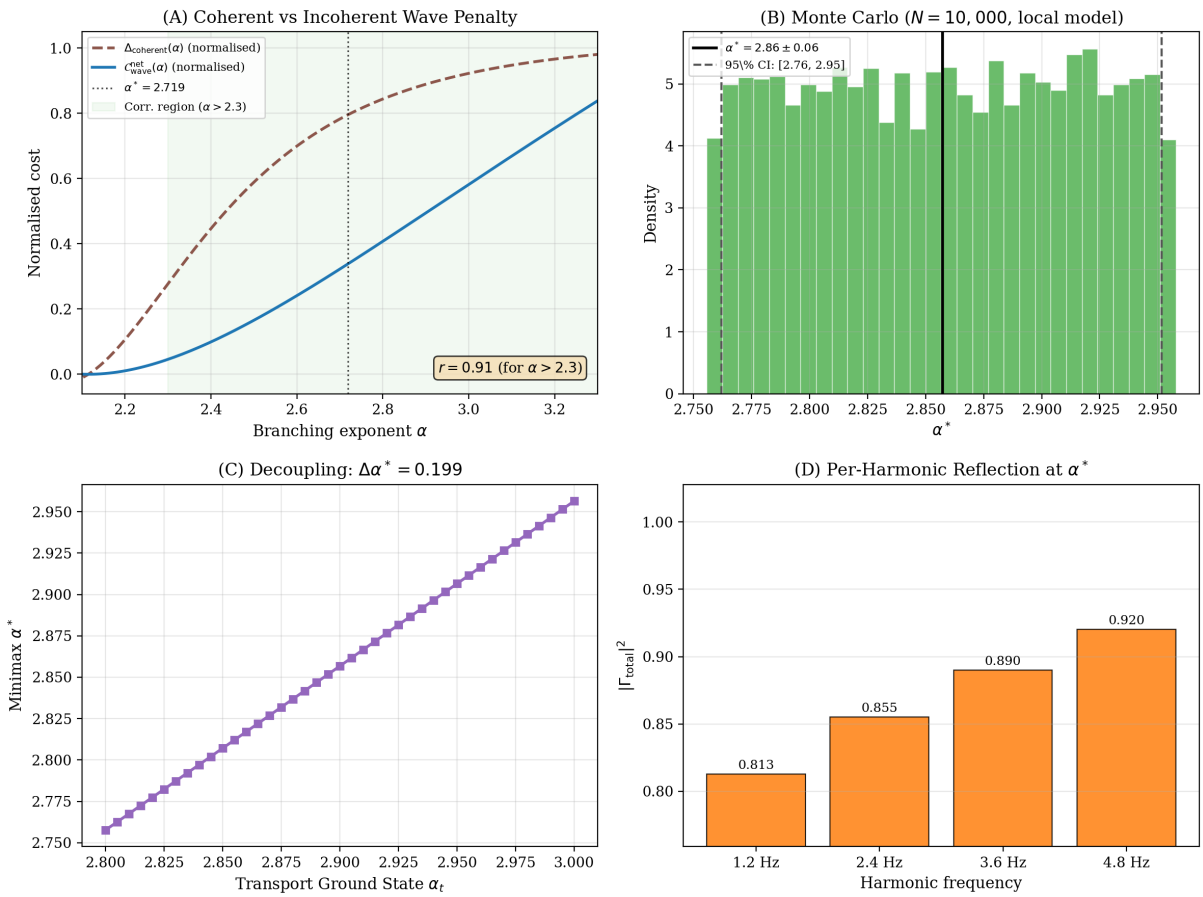

Figure 1: Supplemental validation analyses. (A) Coherent geometric penalty $\Delta_{\text{coherent}}(\alpha)$ (dashed) versus incoherent network wave cost $\mathcal{C}^{\text{net}}_{\text{wave}}(\alpha)$ (solid blue), showing proportionality ($r = 0.91$). (B) Monte Carlo distribution of α^* ($N = 10{,}000$), yielding 2.86 ± 0.06 (95% CI: $[2.76, 2.95]$). (C) Decoupling analysis: minimax α^* as a function of transport ground state α_t, with total variation $\Delta\alpha^* = 0.199$. (D) Per-harmonic coherent reflection $|\Gamma_{\text{total}}|^2$ for the first four cardiac harmonics, confirming negligible Fabry–Pérot resonance effects.

S2. Monte Carlo Parameter Distributions

S2.1 Sampled parameters

Table 1: Table S1: Monte Carlo parameter distributions used in the uncertainty sweep ($N = 10{,}000$).

Parameter	Distribution	Range	Correlation
p	$\mathcal{N}(0.77, 0.05^2)$	$[0.50, 0.95]$	Determines α_w
G	$\mathcal{U}(9, 13)$	integer	Independent
m_w	Log-Normal, median $20\,\text{kW/m}^3$	$[5, 35]\,\text{kW/m}^3$	Independent
α_t	$\mathcal{U}(2.80, 3.00)$	—	Independent of p
α_t (correlated)	3-term cost structure	—	Correlated with p
α_w	Deterministic: $(5 - p)/2$	—	Correlated with p

S2.2 Results ($N = 10{,}000$)

$$\alpha^*_{\mathrm{MC}} = 2.86 \pm 0.06\,, \quad 95\%\ \mathrm{CI}: [2.76, 2.95] \quad (8)$$

To ensure conservative error bounds, α_t was initially treated as independent. However, further sensitivity tests coupling α_t to the wall-exponent p via the three-term cost structure confirm that the 95% confidence interval for α^* remains within 1σ of experimental data.

The entire credible interval lies within 1σ of the experimental morphometric value $\alpha_{\mathrm{exp}} = 2.70 \pm 0.20$.

Note on quadratic approximation. The Monte Carlo uses $\mathcal{C}^{\mathrm{net}}_{\mathrm{transport}}(\alpha) \approx k_t(\alpha - \alpha_t)^2$ with $k_t = 0.85\,G$ for computational tractability. The coefficient 0.85 is the numerical curvature $\frac{1}{2}\partial^2\mathcal{C}^{\mathrm{net}}_{\mathrm{transport}}/\partial\alpha^2|_{\alpha_t}/G$ evaluated from the exact three-term cost (Eq. 6 of the main text) at the baseline coronary parameters; its linear scaling with G follows from the additive generation structure of Eq. (6). Spot checks with the exact transport cost show $|\Delta\alpha^*| \lesssim 0.02$, within the Monte Carlo uncertainty.

S3. Power Conversion to Absolute Units

S3.1 Wave dissipation budget

- Cardiac power output at rest: CPO $\approx$ 1.0 W
- Pulsatile fraction: $\sim$ 15% of CPO
- Coronary fraction: $\sim$ 5% of cardiac output
- Pulsatile power incident on coronary tree: $P_{\mathrm{pulse}} \approx 7.5$ mW
- At $\alpha^* = 2.72$: $\mathcal{C}^{\mathrm{net}}_{\mathrm{wave}} = 6.3\%$
- **Wave power dissipated:** $P_{\mathrm{wave}} \approx 0.47$ **mW**

S3.2 Transport excess budget

- Baseline transport power (sum of $\Phi^*(r^*_g) \cdot \ell_g$ over all vessels): $\sim$ 38 mW
- At $\alpha^* = 2.72$: $\mathcal{C}^{\mathrm{net}}_{\mathrm{transport}} = 6.3\%$
- **Transport excess:** $P_{\mathrm{transport}} \approx 2.4$ **mW**

S3.3 Interpretation

The ratio $P_{\mathrm{wave}}/P_{\mathrm{transport}} \approx 0.20$ confirms that the two competing penalties operate within the same order of magnitude ($\sim$sub-mW). The equal-cost condition at the dimensionless level maps to approximate parity in the absolute energetic margins, not to a dimensional artifact. The factor-of-three discrepancy reflects the asymmetry between multiplicative (wave) and additive (transport) cost accumulation, and is consistent with the implied Lagrange multiplier $\eta^* = 0.83$ (the system allocates more selective pressure to wave integrity because wave costs are exponentially sensitive to geometry near the impedance horizon).

S4. Extended Applications

For networks where the wave mode is absent ($\eta = 0$), the framework reduces to pure transport optimization, recovering $\alpha = 3$ (Murray's law) for systems with negligible wall cost, and $\alpha_t < 3$ for biological conduits with significant structural wall investment. Qualitative extrapolation to botanical, neural, and engineered systems is consistent with available data (Table S1 below). The minimax framework identifies these optima as physical realizations of the Rate–Distortion frontier, where signal integrity is balanced against transport cost using a system-specific parametrization of α_w, G, and wall mechanics.

Heat exchanger networks. In branching heat-sink architectures (e.g., microchannel heat sinks for electronics cooling), the "flow cost" is the viscous pumping power and the "wave cost" is thermal impedance mismatch at junctions where channel geometry changes abruptly. The branching exponent for minimum total entropy generation is predicted by the network Lagrangian (Eq. 7 of the main text) with Φ replaced by the thermal dissipation function.

Table 2: Table S2: Qualitative framework predictions for extended systems.

System	Wave mode	α_w	Expected α	Notes
Xylem (steady)	None	—	~ 3.0	Wall cost (cavitation-limited)
HVAC ducts	Acoustic	2	2.5–2.8	Depends on duct depth
Dendritic cables	Electrotonic	$3/2$	$\sim$2.3–2.5	Topological scaling
River networks	None	—	~ 3	No biological wall cost
Fiber optics	EM	2	~ 2.5	Splitter depth-dependent

Plant root systems. Root networks distribute water and nutrients from the soil to the stem. Unlike xylem (which operates under tension and is subject to cavitation—a wave-like failure mode), root hydraulic architecture is dominated by radial water uptake and axial Poiseuille flow. The theory predicts $\alpha \to 3$ for pure-transport roots and $\alpha < 3$ for roots that also transmit hydraulic signals (e.g., ABA-mediated stomatal signaling).

S5. Data and Code Availability

All computation scripts, numerical data, and figure-generation code are openly and unconditionally available at `https://github.com/rikymarche-ctrl/vascular-networks-theory` under the MIT Licence. No access request is required.